

EXERCISES

QUESTION 1:

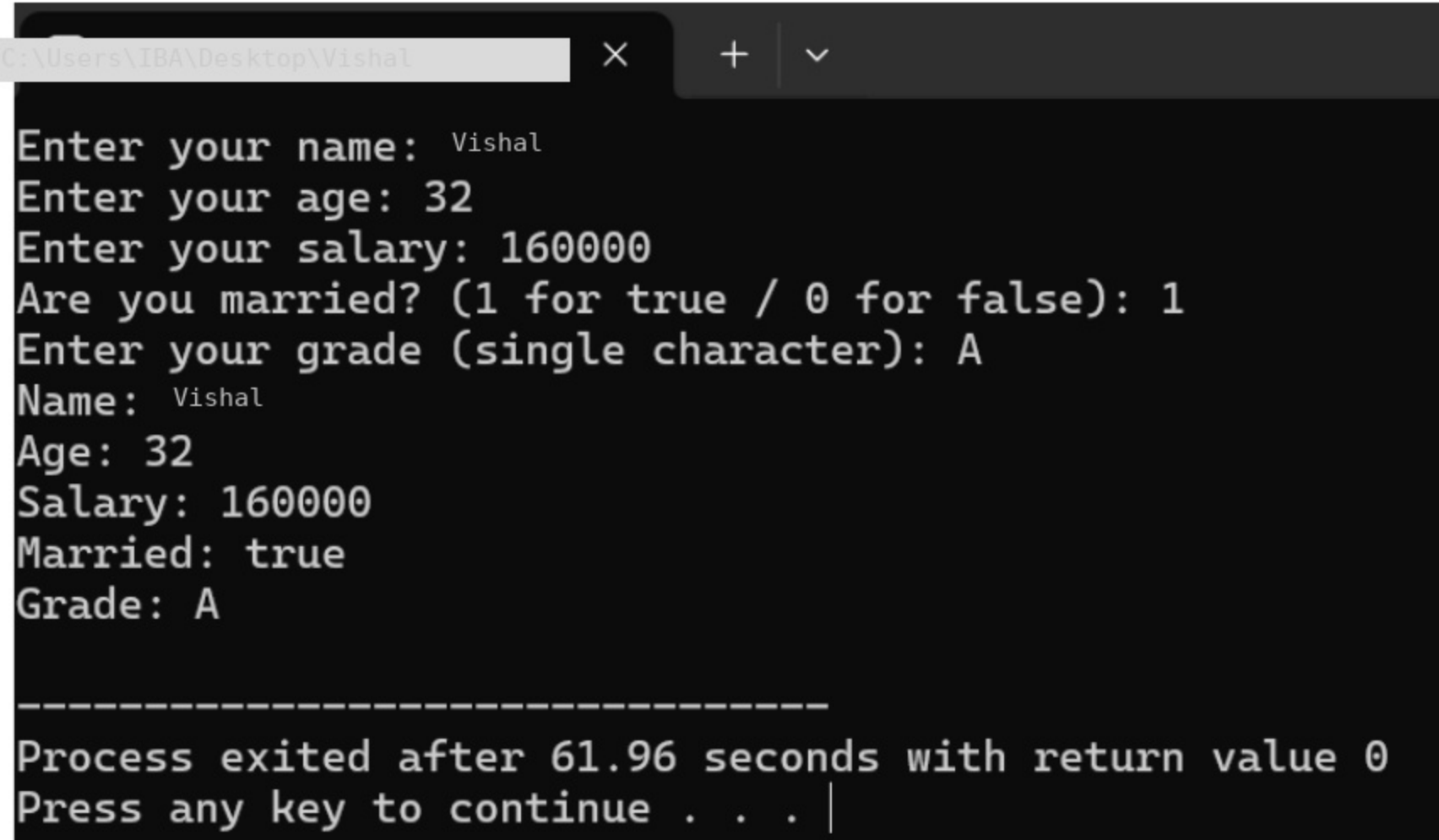
Write a program that takes 5 values from the user:

- ✓ one value of String (name),
- ✓ one value of int (age),
- ✓ one value of double (salary),
- ✓ one value of boolean (isMarried),
- ✓ one value of char (grade).
- ✓ Print each value on a separate line.

CODE:

```
#include <iostream>
using namespace std;
int main() {
    string name;
    int age;
    double salary;
    bool isMarried;
    char grade;
    cout << "Enter your name: ";
    getline(cin, name);
    cout << "Enter your age: ";
    cin >> age;
    cout << "Enter your salary: ";
    cin >> salary;
    cout << "Are you married? (1 for true / 0 for false): ";
    cin >> isMarried;
    cout << "Enter your grade (single character): ";
    cin >> grade;
    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Salary: " << salary << endl;
    cout << "Married: " << boolalpha << isMarried << endl;
    cout << "Grade: " << grade << endl;
    return 0;}
```

OUTPUT:



```
Enter your name: Vishal
Enter your age: 32
Enter your salary: 160000
Are you married? (1 for true / 0 for false): 1
Enter your grade (single character): A
Name: Vishal
Age: 32
Salary: 160000
Married: true
Grade: A

-----
Process exited after 61.96 seconds with return value 0
Press any key to continue . . . |
```

QUESTION 2:

Write a program that takes Marks Obtained (float) and Total Marks (int) from the user. Calculate and print **Percentage** using the following formula:

$$\text{Percentage} = \frac{\text{MarksObtained}}{\text{TotalMarks}} \times 100$$

CODE:

```
#include <iostream>
using namespace std;

int main() {
    float marksObtained;
    int totalMarks;

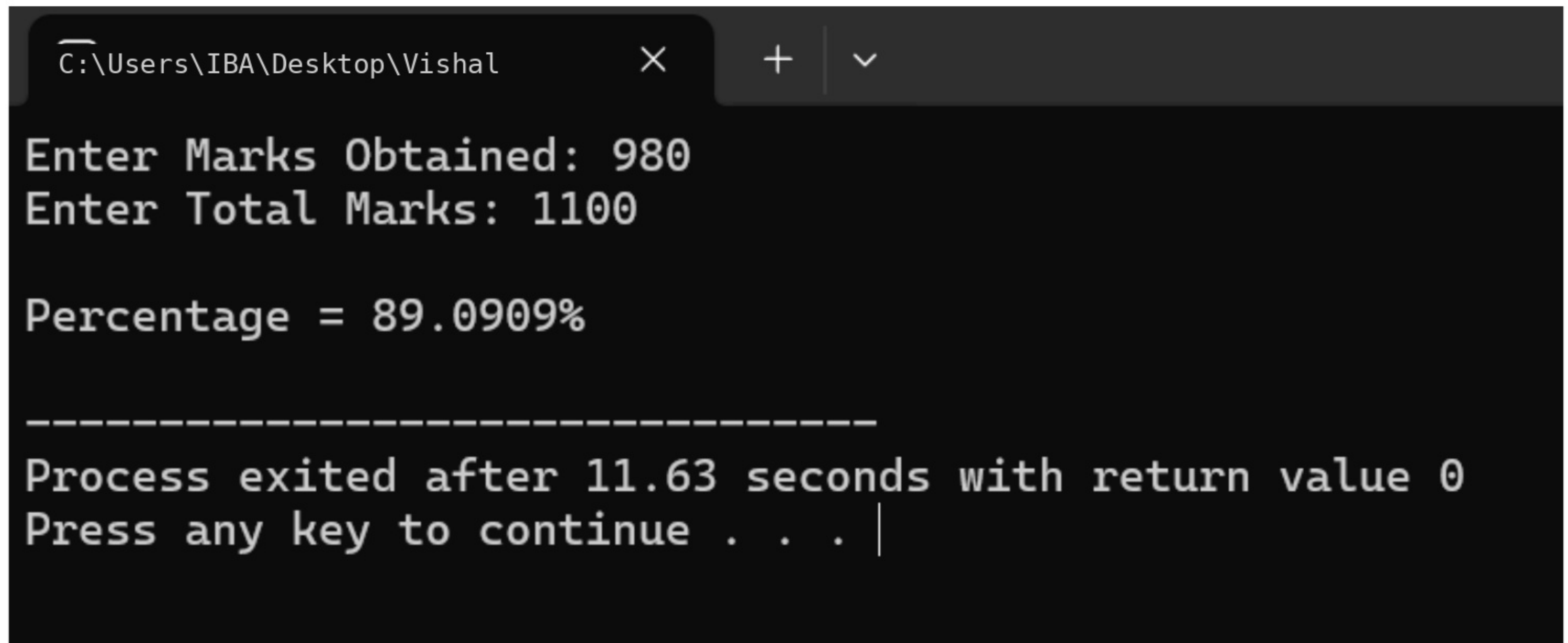
    cout << "Enter Marks Obtained: ";
    cin >> marksObtained;
    cout << "Enter Total Marks: ";
    cin >> totalMarks;

    float percentage = (marksObtained / totalMarks) * 100;

    cout << "\nPercentage = " << percentage << "%" << endl;

    return 0;
}
```

OUTPUT:



```
C:\Users\IBA\Desktop\Vishal

Enter Marks Obtained: 980
Enter Total Marks: 1100

Percentage = 89.0909%

-----
Process exited after 11.63 seconds with return value 0
Press any key to continue . . . |
```


QUESTION 3:

Write a program that finds the value of X by using given formula. Take value of 'm' and 'n' from user:

$$X = (m^2 + n^2) - 2mn$$

CODE:

```
#include <iostream>
using namespace std;

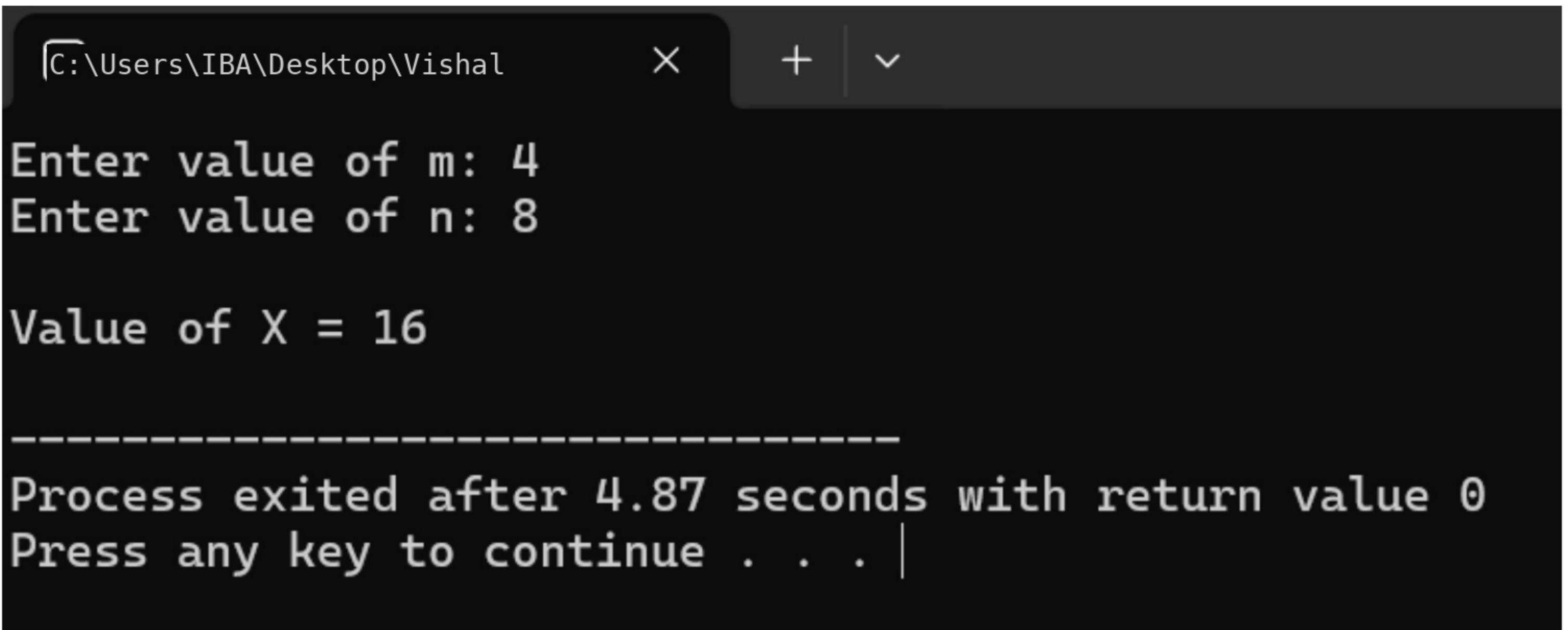
int main() {
    int m, n, X;

    cout << "Enter value of m: ";
    cin >> m;
    cout << "Enter value of n: ";
    cin >> n;

    X = (m * m + n * n) - (2 * m * n);

    cout << "\nValue of X = " << X << endl;

    return 0;
}
```

OUTPUT:

```
C:\Users\IBA\Desktop\Vishal

Enter value of m: 4
Enter value of n: 8

Value of X = 16

-----
Process exited after 4.87 seconds with return value 0
Press any key to continue . . . |
```

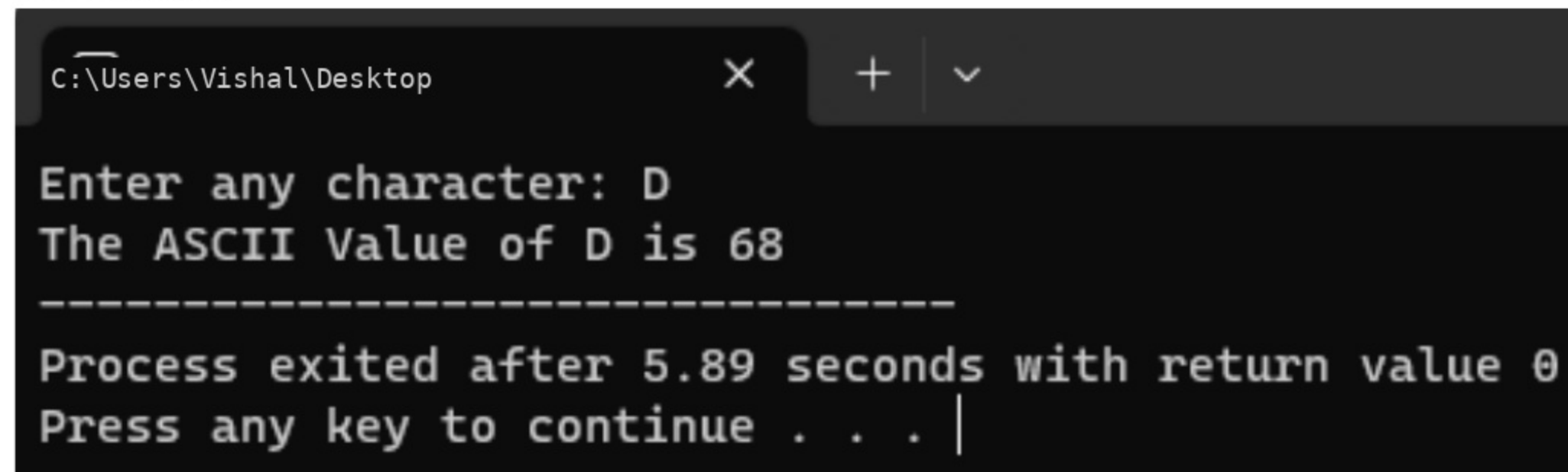
QUESTION 4:

Write a program that asks the user to type character then the program will generate its ASCII value. The sample output is shown below:

```
Enter any character: z
The ASCII Value of z is 122
```

CODE:

```
#include<iostream>
using namespace std;
int main()
{
    char a;
    int b;
    cout << "Enter any character: ";
    cin >> a;
    b=a;
    cout << "The ASCII Value of " << a << " is " << b;
    return 0;
}
```

OUTPUT:

The screenshot shows a Windows command prompt window with the title bar 'C:\Users\Vishal\Desktop'. The window contains the following text:

```
Enter any character: D
The ASCII Value of D is 68
-----
Process exited after 5.89 seconds with return value 0
Press any key to continue . . . |
```


QUESTION 5:

Write a program that takes a Number (integer) input from the user and then it prints the Character associated with that ASCII. Let's suppose if User input is 65 then A should be printed on screen.

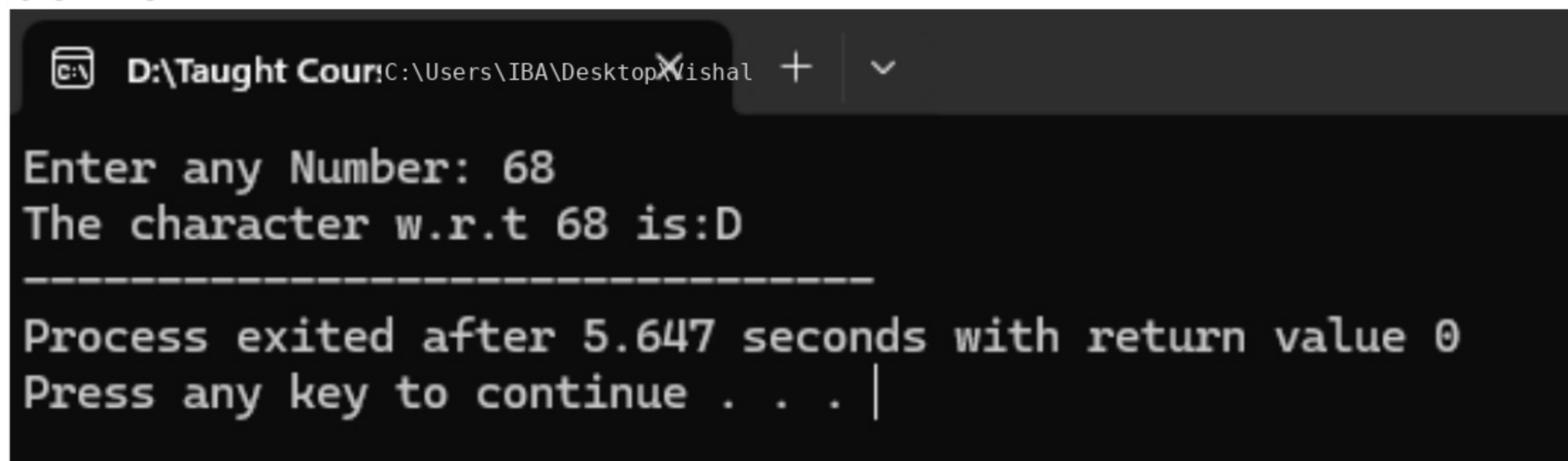
Enter any Number: 65

The character w.r.t 65 is A

CODE:

```
#include<iostream>
using namespace std;
int main()
{
    int a;
    cout<<"Enter any Number: ";
    cin>>a;
    cout<<"The character w.r.t "<<a<<" is:"<<char(a);
return 0;
}
```

OUTPUT:

A screenshot of a Windows command prompt window. The title bar shows the file path 'D:\Taught Cour...' and the file name 'C:\Users\IBA\Desktop\Wishal'. The command prompt displays the program's output: 'Enter any Number: 68' followed by 'The character w.r.t 68 is:D'. Below this, a separator line of dashes is shown, followed by the message 'Process exited after 5.647 seconds with return value 0' and 'Press any key to continue . . . |'.

QUESTION 6:

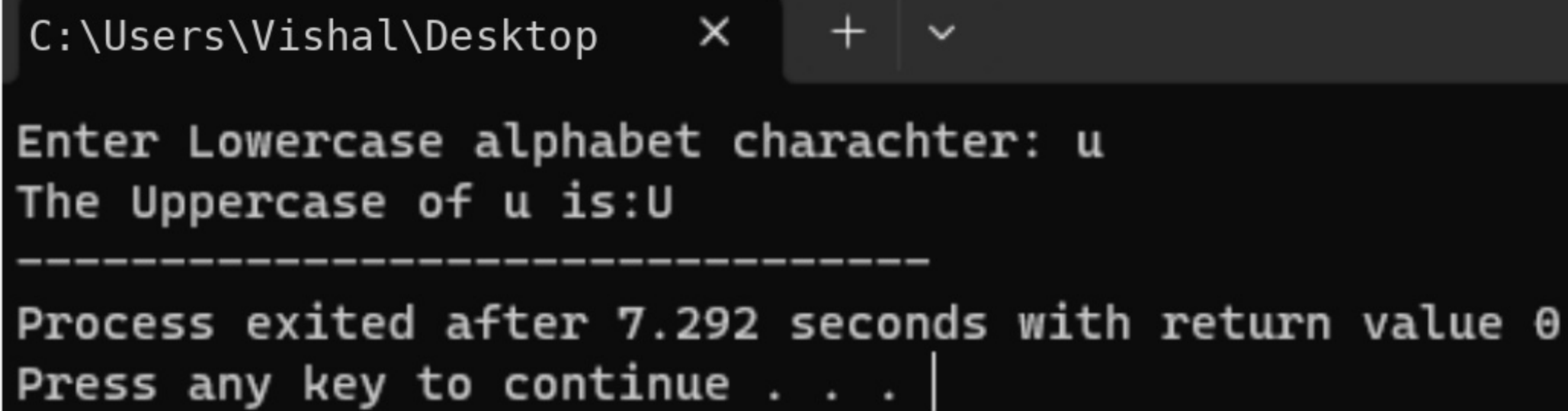
Write a program that asks the user to type lowercase (small) alphabet character then the program will generate its uppercase (capital) alphabet character. The sample output is shown below:

```
Enter Lowercase alphabet character: m
The Uppercase of m is:M
```

CODE:

```
#include<iostream>
using namespace std;
int main()
{
    char a;
    cout<<"Enter Lowercase alphabet character: ";
    cin>>a;
    cout<<"The Uppercase of "<<a<<" is:"<<char(a-32);

    return 0;
}
```

OUTPUT:

```
C:\Users\Vishal\Desktop  X  +  v
Enter Lowercase alphabet character: u
The Uppercase of u is:U
-----
Process exited after 7.292 seconds with return value 0
Press any key to continue . . . |
```

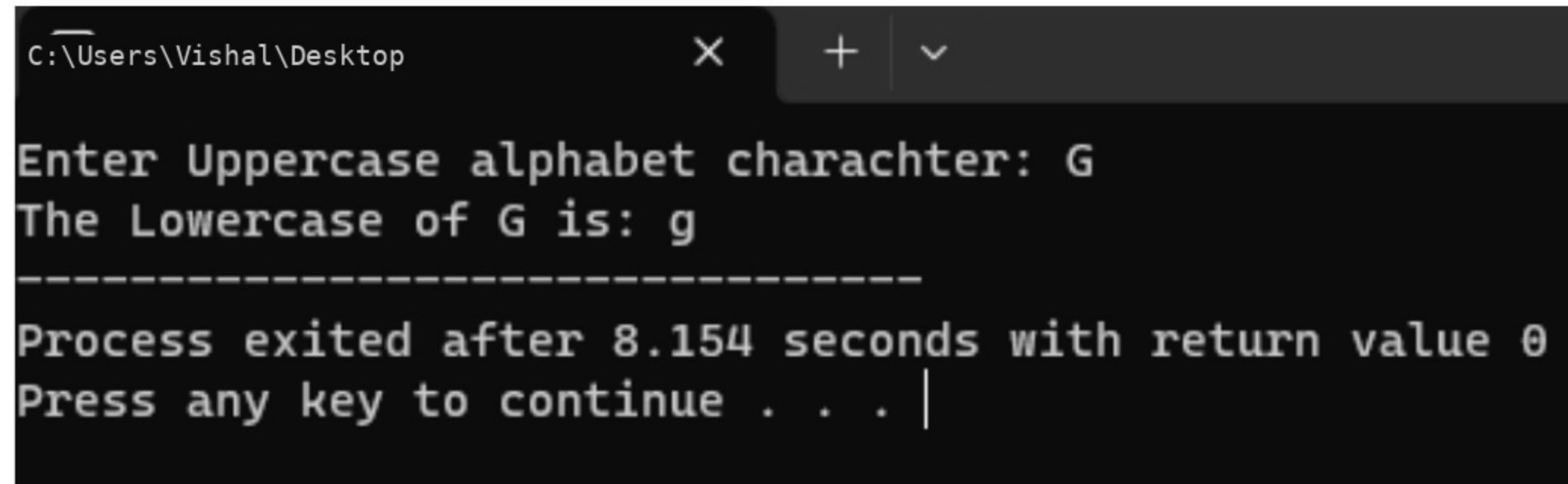

QUESTION 7:

Write a program that asks the user to type uppercase (capital) alphabet character then the program will generate its lowercase (small) alphabet character. The sample output is shown below:

```
Enter Uppercase alphabet character: B
The Lowercase of B is: b
```

CODE:

```
#include<iostream>
using namespace std;
int main()
{
    char a;
    cout<<"Enter Uppercase alphabet character: ";
    cin>>a;
    cout<<"The Lowercase of "<<a<<" is: "<<char(a+32);
    return 0;
}
```

OUTPUT:

```
C:\Users\Vishal\Desktop
Enter Uppercase alphabet character: G
The Lowercase of G is: g
-----
Process exited after 8.154 seconds with return value 0
Press any key to continue . . . |
```